BRAND POPULARITY AND CUSTOMER PREFERENCE ANALYSIS

EDAV – Python Course-End Project Report

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Abstract

This project applies Python-based exploratory data analysis to real online retail transactions from the UCI Machine Learning Repository. It studies product popularity using quantities and transaction frequency, revenue by product and country, monthly revenue trends, and repeat purchasing among identified customers. Pandas and NumPy are used for data preparation and calculations, while Matplotlib and Seaborn are used for visualizations.

1. Introduction

Understanding what customers buy and how purchasing patterns change is useful for business planning. This project applies exploratory data analysis to transaction records to summarize product-level sales, geographic revenue, and repeat-order behavior.

2. Problem Statement

Retail transaction data can be large and difficult to interpret manually. This project organizes the data and uses visual and statistical summaries to identify patterns that may inform product and customer analysis.

3. Objectives

• Identify products with the highest units sold and transaction frequency.

• Compare product revenue.

• Analyze revenue across countries and months.

• Estimate repeat purchasing among customers with recorded IDs.

• Communicate findings through understandable charts.

4. Dataset Description

Dataset: UCI Online Retail, created from transactions of a UK-based non-store online retailer between 1 December 2010 and 9 December 2011. The original dataset contains 541,909 rows and fields including invoice number, stock code, product description, quantity, invoice date, unit price, customer ID, and country. Source: https://archive.ics.uci.edu/dataset/352/online+retail

5. Important Scope Note

The dataset does not provide explicit brand names or customer satisfaction ratings. Therefore, product descriptions are used as a proxy for product popularity, and purchasing behavior is analyzed instead of claiming survey-based customer preference or satisfaction. Sales patterns do not prove why a customer selected a product.

6. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, and GitHub. The openpyxl package reads the Excel dataset, and ucimlrepo can attempt to retrieve the dataset programmatically.

7. Methodology

1. Load the UCI dataset from a local Excel/CSV file or attempt programmatic retrieval.

2. Inspect columns, missing values, and data types.

3. Parse dates and numeric fields.

4. Exclude cancelled invoices, negative/zero quantities, and non-positive prices from positive-sales analysis.

5. Calculate revenue as Quantity × UnitPrice.

6. Group records by product, country, and month.

7. Create bar charts and a monthly trend line.

8. Save summary tables and metrics for the final findings section.

8. Visualizations

The analysis script generates four charts: top products by units sold, top products by revenue, top countries by revenue, and monthly revenue trend. Insert the generated PNG files from the visualizations folder here after running the code.

9. Results and Discussion

Complete this section after running the project. Use report/analysis_summary.txt and the CSV tables to state the actual top products, revenue totals, date range, and repeat-customer share. Do not invent results or copy expected values without verifying the output.

10. Limitations

The dataset is historical and relates to one retailer. It does not contain explicit brand labels, customer ratings, stated reasons for purchase, or complete customer identifiers for every transaction. Revenue and quantity are different measures of popularity, and results may not generalize to all markets.

11. Conclusion

This project demonstrates the use of Python for practical exploratory data analysis. It transforms raw retail transactions into summaries and visualizations that reveal product sales patterns, country-level revenue, monthly trends, and repeat purchasing among identified customers. The project can be extended with customer surveys or review data containing actual brands and satisfaction ratings.

12. Future Scope

Add a survey dataset with brand labels and customer ratings; analyze customer reviews using natural language processing; build an interactive dashboard; and compare purchasing patterns over longer time periods.

13. References

Chen, D. (2015). Online Retail [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5BW33

Pandas documentation: https://pandas.pydata.org/docs/

NumPy documentation: https://numpy.org/doc/

Matplotlib documentation: https://matplotlib.org/stable/

Seaborn documentation: https://seaborn.pydata.org/

GitHub documentation: https://docs.github.com/